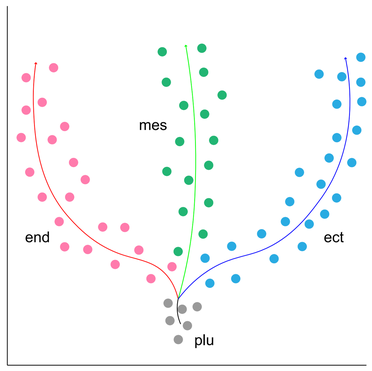
K-Spheres: A Novel Methodology for Clustering Differentiating Stem Cells

Aditya Iyer[1], Adriana Beltran[2], Tarek Zikry[1], Sam Wolff[2], Jamshaid Shahir[2], Didong Li[1]*, Jeremy Purvis[2]*

[1] Dept. of Biostatistics, University of North Carolina at Chapel Hill
[2] Dept. of Genetics, University of North Carolina at Chapel Hill
* Senior Corresponding Author(s)

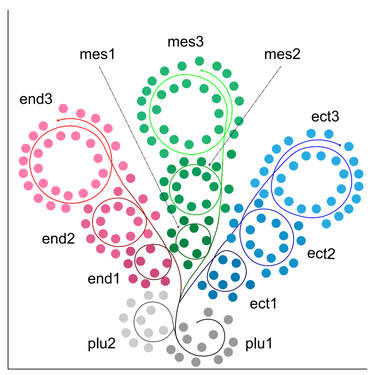
Background: Cyclic vs. Acyclic Differentiation Trajectories

Naïve Approach: Acyclic Differentiation Trajectories



Abbreviation	Lineage
plu	Pluripotent Cells
end	Endoderm Cells
mes	Mesoderm Cells
ect	Ectoderm Cells

Sophisticated Approach: Cyclic Differentiation Trajectories



Abbreviation	Lineage
plu1	Pluripotent Population 1
plu2	Pluripotent Population 2
end1	Early Endoderm Cells
end2	Middle Endoderm Cells
end3	Late Endoderm Cells
mes1	Early Mesoderm Cells
mes2	Middle Mesoderm Cells
mes3	Late Mesoderm Cells
ect1	Early Ectoderm Cells
ect2	Middle Ectoderm Cells
ect3	Late Ectoderm Cells

- Key goal: understand the molecular mechanisms by which human embryonic stem cells differentiate into germ layers.
- Challenge: stem cells simultaneously undergo two processes,
  1. differentiation, sequential changes in the molecular states of cells during development
  2. the cell cycle, a cyclical process by which one cell gives rise to two daughter cells.
- Hypothesis: given the interdependence between the two processes, pluripotent cells progress through differentiation via a series of cyclical molecular states.
- Each cyclical state may correspond to a germ layer lineage (or sub-trajectory such as early, middle, or late endoderm) or a population of pluripotent cells.
- Trajectory inference methods do not account for our hypothesis.

References
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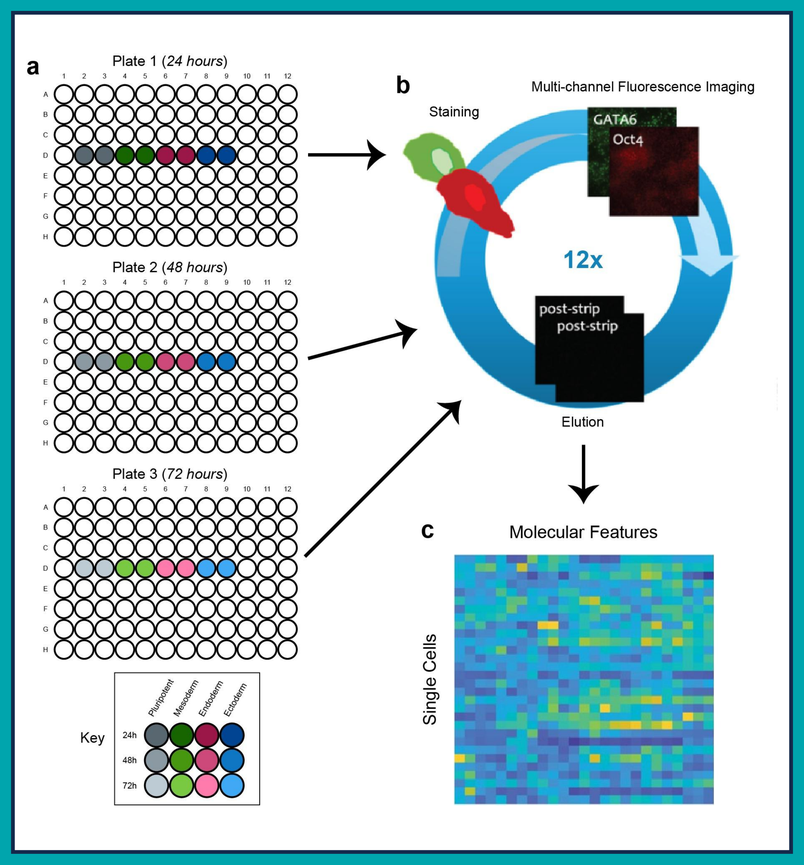
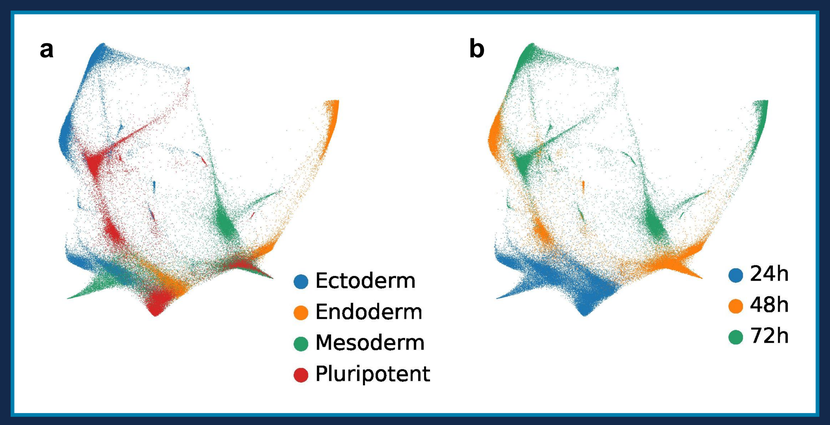
Experimental Procedure



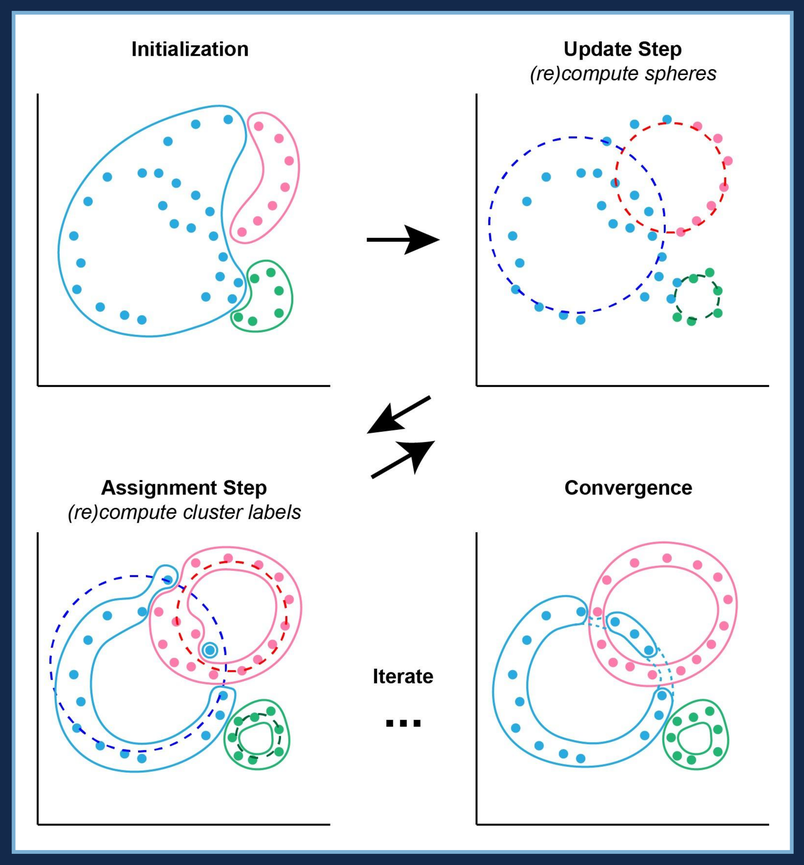
a. We seeded human embryonic stem cells to differentiate into one of the three germ layers: Endoderm, Mesoderm, or Ectoderm.
   - We also seeded cells to remain pluripotent as a control.
   - Cells were left to differentiate for 24, 48, or 72 hours.
   - We must note that, while we seeded populations of cells to differentiate into prespecified lineages, they likely contain impurities due to the intractability of directed differentiation.
b. We performed 12 rounds of Iterative Indirect Immunofluorescence Imaging (4i)[1], a novel multiplexed microscopy technique, for a curated set of proteomic biomarkers associated with differentiation and the cell cycle.
c. The 4i data underwent a series of preprocessing steps:
   - image segmentation to identify cell nuclei
   - the subsampling of cells for computational feasibility
   - the removal of proteomic features with poor antibody staining.
   - Final dataset: 20 z-normalized fluorescence intensity measurements (features) for 109,826 cells (observations).

Low-Dimensional Manifold Embeddings with PHATE



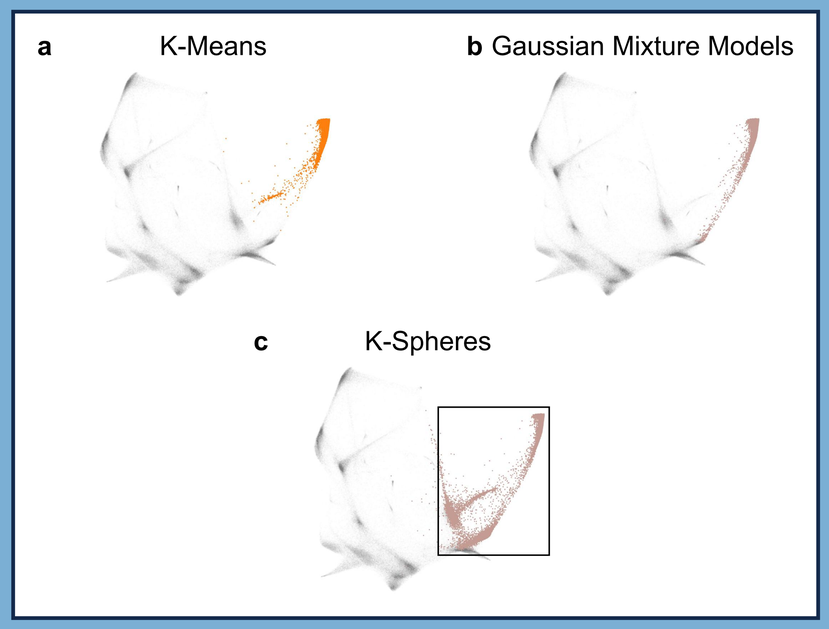
- For visualization, we generated 2D embeddings of the 4i data using PHATE (Potential of Heat diffusion for Affinity-based Transition Embedding)[2].
  - We produced overlays of the intended differentiation lineage (a) and the length of differentiation (b).
- Neither overlay contains a category that encompasses a complete cyclic structure, which contradicts our hypothesis.
  - The absence of cyclic structures supports our assumption that several of our seeded cells did not differentiate as intended.
  - We must therefore retrospectively identify the lineage to which each cell belongs.
  - K-spheres can help identify germ layer differentiation trajectories.

Formulation of K-Spheres



- Goal: partition data into subsets that exhibit spherical patterns.
- Algorithm: analogous to k-means clustering but evaluates distances to low-dimensional spheres instead of cluster centers.
  - Low dimensional spherical embeddings are computed using spherical component analysis (SPCA)[3].
  - SPCA is parameterized by a center, radius, and low-dimensional affine subspace.
- K-spheres captures the geometry underlying differentation and the cell cycle in a biologically interpretable manner.
  - Shifts in cluster center and radius can inform us of changes in biomolecular content and brevity of a cell cycle, respectively.
- K-spheres is part of a family of "cycle-based" clustering methods, which includes mixtures of Fisher–Gaussian (FG) kernels[4].
  - Mixtures of FG kernels are a probabilistic ("soft clustering") generalization of k-spheres.
  - Similar to how Gaussian mixture models are a probabilistic generalization of k-means.

Results: Endoderm Identification



- K-means (a), Gaussian mixture models (b), and k-spheres (c) each identified clusters comprised mostly of endoderm cells.
- Only the k-spheres cluster contains a complete cyclic structure.
  - Given our hypothesis, only the k-spheres cluster is likely to encapsulates the near entirety of the endoderm lineage.
  - The k-spheres cluster spans all proliferative cell cycle phases.
  - Endoderm classification was based on the SOX17, EOMES, and GATA6 differentiation markers.